CARIINKERJA.ID · OPERATOR GUIDE

Skill Map — how it works

Public pages that show which skills employers ask for most, per job title, counted from the jobs we already crawl. This guide covers where the numbers come from, how to add a page, and what to check when a page doesn't show.

Written 1 Nov 2026 against `main` @ 4842638. Re-render:
`weasyprint _docs/skill-map.html _docs/skill-map.pdf`

1 What it is

`cariinkerja.id/skill/` lists every live page. Each page, e.g. `/skill/data-analyst/`, shows for one job title:

- top **15 hard skills** and top **8 soft skills**, each with the % of postings that ask for it;
- the remote / on-site / hybrid mix and the job-type mix;
- the date window and the job count behind the numbers;
- a CTA with the price, linking to `/login?ref=skillmap-<slug>`.

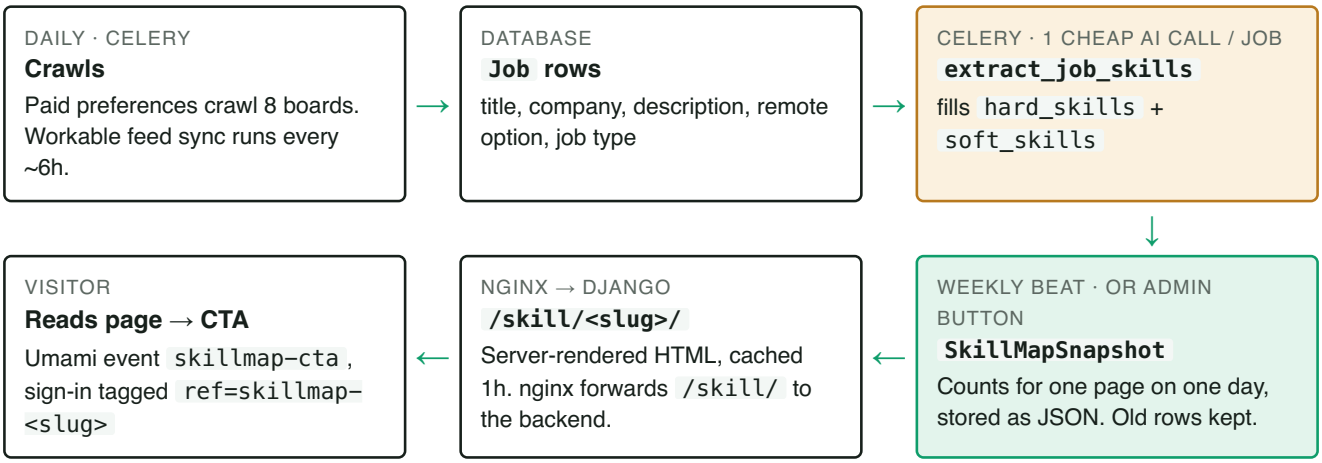
No login. It's a top-of-funnel tool for search and social posts. It shows the market, never the visitor's own score, so it is not a free tier.

Rules that never bend

- **Only aggregate counts go public.** No postings, no company names, no JobPosting schema. Himalayas' API terms forbid passing its jobs on.
- **Every number is dated.** Each page states its window and job count.
- **Thin pages stay hidden.** Under `min_jobs` (default 30) jobs, the page returns 404 and is left out of the sitemap.
- Copy is casual Bahasa ("kamu"), says "AI", never a model name.

2 Where the numbers come from

Nothing new gets crawled for Skill Map. It reuses the skills the pipeline already extracts from every job.



Two things to keep in mind:

- **The page shows the latest snapshot, not live data.** Between refreshes, the numbers don't move.
- **A job without extracted skills is invisible to Skill Map.** Jobs touched by a preference crawl get skills automatically. Workable-synced jobs don't until you backfill them (step 2 below).

3 Adding a page, step by step

1 Find titles worth a page

```
# defaults shown
uv run manage.py skill_map_candidates --limit 40 --min-jobs 30 --window-days 90
```

Lists the job titles users search most, and how many jobs a page for that title would count today. Example output (made-up numbers):

```
searches   jobs  ok  title  [slug]
      14    212  ✓   Data Analyst  [data-analyst] (page exists)
       9     87  ✓   Frontend Developer  [frontend-developer]
       6     12      UI UX Designer  [ui-ux-designer]
```

COLUMN	MEANING
`searches`	How many preferences use this title. Case and extra spaces are merged ("data analyst" = "Data Analyst"). Counts **every** preference, including unpaid and expired ones.
`jobs`	Jobs a page with this title would count right now, using the same matching as the real page (section 4).
`ok`	✓ when `jobs` ≥ `--min-jobs`, so the page would be visible once published.
`title`	The most common spelling users typed.
`[slug]`	The URL slug the page would get. `(page exists)` = you already made it.

Pick: high `searches` + ✓ + no page yet.

2 Optional: give Workable jobs their skills

```
# see how many jobs (= AI calls) first, then run for real
uv run manage.py backfill_job_skills --source workable --since-days 90 --dry-run
uv run manage.py backfill_job_skills --source workable --since-days 90
```

Queues one cheap AI call per job to Celery (add `--sync` to run inline without a worker). Re-run `skill_map_candidates` afterwards: the `jobs` counts go up.

3 Create the page in Django admin

Admin → Jobs → Skill map pages → Add.

FIELD	WHAT TO PUT
`title`	Shown on the page, e.g. “Data Analyst”.
`slug`	Auto-filled from the title. Becomes `/skill/<slug>/`. Don’t change it after publishing, or old links break.
`match_words`	Leave empty to match on the title. Fill it when the title is too narrow or too wide (e.g. title “Frontend Developer”, match words “frontend”).
`exclude_words`	Any job title containing one of these is dropped, e.g. “senior lead manager” for an entry-level page.
`min_jobs`	Default 30. Below this the page stays hidden.
`is_published`	Leave off until you’ve checked the count in step 4.

4 Compute the first snapshot

In the page list, tick the page → action **“Refresh snapshot now”**. The message shows the job count, and the **jobs (latest)** column updates. Too few jobs? Loosen `match_words` / `exclude_words` and refresh again.

5 Publish

Tick `is_published` and save. The page and the `/skill/` index are cached for 1 hour, so a new page can take up to an hour to appear.

6 One-time setup: the weekly refresh

Admin → Periodic tasks → Add: task `jobs.tasks.refresh_skill_maps`, weekly. It refreshes **every** page, published or not. Without it, pages keep showing their first snapshot forever.

4 How the numbers are counted

All of this lives in `jobs/skill_map.py`. For one page, on refresh:

a. Which jobs count

1. **Window:** jobs added to our database in the last **90 days** (by when we crawled them, not when they were posted).
2. **Has skills:** jobs with no extracted hard skills are skipped.
3. **Title match:** every match word must *start* a word in the job title. Case doesn’t matter.

MATCH WORDS	JOB TITLE	COUNTED?
analyst	Data Analysts	**yes — “analyst” starts “Analysts”**
data analyst	Analyst, Data & Insights	**yes — word order doesn’t matter**
ui	Build Engineer	**no — “ui” is inside “Build”, not at its start**
frontend	Front End Developer	**no — “front” and “end” are separate words**
analyst (exclude: senior)	Senior Data Analyst	**no — excluded word present**

4. **Duplicates:** the same company + same title from two job boards counts once.

b. How skills are ranked

- **Once per job.** A posting that says “SQL” five times still adds 1 to SQL.
- **Spellings merge.** “SQL”, “sql” and “ SQL ” are one skill. Fixed aliases also merge: `ms excel / microsoft excel → excel` · `ms office → microsoft office` · `reactjs / react.js → react` · `nodejs / node → node.js` · `vuejs / vue → vue.js` · `postgres → postgresql` · `golang → go` · `js → javascript` · `ts → typescript`. The page shows the most common original spelling.
- **% = jobs that ask for the skill ÷ all counted jobs.** 60% SQL means 6 in 10 postings mention SQL. The percentages don’t add up to 100.
- Top 15 hard and top 8 soft are kept. Ties are sorted alphabetically.

c. Work-style mix

Remote / on-site / hybrid and full-time / part-time / contract / internship are shares of the same counted jobs. Many postings don’t state them, so these don’t add up to 100% either. The page says “Sisanya ga nyebut di lokernya” for that reason.

d. Snapshots

One snapshot per page per day. Refreshing twice on the same day overwrites that day’s row. Older snapshots are kept (for trends later). The page always shows the newest one.

5 When something looks wrong

SYMPTOM	CAUSE → FIX
Page gives 404	`is_published` is off, there is no snapshot yet, **or** the newest snapshot has fewer than `min_jobs` jobs. Check the **jobs (latest)** column. A page can disappear on its own after a weekly refresh if jobs dried up.
Page live but numbers old	Weekly beat task not registered (step 6), or the 1-hour cache. Run “Refresh snapshot now” and wait up to an hour.
Fewer jobs than `skill_map_candidates` said	The command matches on the bare title only. It ignores your `match_words` and `exclude_words`. Trust the admin refresh count.
Two names for one skill	e.g. “Power BI” vs “PowerBI”. Add an alias to `SKILL_ALIASES` in `jobs/skill_map.py` (a code change + deploy), then refresh.
Job count lower than expected	Jobs without skills don’t count. Backfill (step 2). Also check spelling variants the title rule misses, like “Front End” vs “Frontend”.
Index page empty	No page is both published and above `min_jobs`.

Don’t

- Add company names, job links, or any single posting to the templates. Aggregates only.
- Rename a published slug. Old links and search rankings break.
- Lower `min_jobs` to force a page live. A page built on 8 jobs gives misleading percentages.

6 Where things live

FILE	WHAT’S THERE
`jobs/skill_map.py`	Matching, ranking, aliases, snapshot write, visibility rules. Constants: `WINDOW_DAYS=90`, `TOP_HARD=15`, `TOP_SOFT=8`.
`jobs/models.py`	`SkillMapPage` (config), `SkillMapSnapshot` (counts as JSON).
`jobs/admin.py`	Page admin, “Refresh snapshot now” action, **jobs (latest)** column.
`jobs/tasks.py`	`refresh_skill_maps` (weekly), `extract_job_skills` (per job).
`jobs/management/commands/`	`skill_map_candidates.py`, `backfill_job_skills.py`.
`jobs/skill_map_views.py` `jobs/skill_map_urls.py`	Index, detail, `sitemap.xml` views. 1-hour cache. Price taken from the cheapest active plan.
`templates/skillmap/`	Page HTML with inline CSS, the CTA and the “Cara kita ngitung” section.
`frontend/nginx.conf`	Forwards `/skill/` from the SPA container to Django. In dev, Vite proxies it to :8000.